

Probability

Type I: Basics of Probability

1. Two dice are rolled, find the probability that the sum is (i) equal to 1 (ii) equal to 4 (iii) less than 13

Solution:

$S =$ Two dice are rolled

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$n(S) = 36$$

- (i) let $A =$ sum is equal to 1

$$A = \{ \quad \} = \phi$$

$$n(A) = 0$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{0}{36} = 0$$

- (ii) let $B =$ sum is equal to 4

$$B = \{(1,3), (2,2), (3,1)\}$$

$$n(B) = 3$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

- (iii) let $C =$ sum is less than 13

$$C = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$n(C) = 36$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{36}{36} = 1$$

- (iv) let $D =$ sum is a prime number

$$D =$$

$$\{(1,1), (1,2), (2,1), (1,4), (2,3), (3,2), (4,1), (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), (5,6), (6,5)\}$$

$$n(D) = 15$$

$$P(D) = \frac{n(D)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$



(v) let $E = \text{sum is an even number}$

$$E = \{(1,1), (1,3), (2,2), (3,1), (1,5), (2,4), (3,3), (4,2), (5,1), (2,6), (3,5), (4,4), (5,3), (6,2), (4,6), (5,5), (6,4), (6,6)\}$$

$$n(E) = 18$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{18}{36} = \frac{1}{2}$$

(vi) let $F = \text{sum is a multiple of 5}$

$$F = \{(1,4), (2,3), (3,2), (4,1), (4,6), (5,5), (6,4)\}$$

$$n(F) = 7$$

$$P(F) = \frac{n(F)}{n(S)} = \frac{7}{36}$$

2. In the play of two dice, a person loses if the sum obtained is 2 or 4 or 12. He wins if the sum is 5 or 11. Find the ratio of his probability of losing to the probability of winning in the first throw

Solution:

$S = \text{two dice are rolled}$

$$n(S) = 6 \times 6 = 6^2 = 36$$

Let $A = \text{losing if sum is 2 or 4 or 12}$

$$A = \{(1,1), (1,3), (2,2), (3,1), (6,6)\}$$

$$n(A) = 5$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36}$$

Let $B = \text{winning if sum is 5 or 11}$

$$B = \{(1,4), (2,3), (3,2), (4,1), (5,6), (6,5)\}$$

$$n(B) = 6$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{6}{36}$$

$$\text{Ratio} = \frac{P(\text{losing})}{P(\text{winning})} = \frac{P(A)}{P(B)} = \frac{5}{6}$$

3. A fair dice is thrown thrice. Find the probability that the sum of the numbers obtained is 10

Solution:

$S = \text{a fair dice thrown thrice}$

$$n(S) = 6 \times 6 \times 6 = 6^3 = 216$$

Outcomes	Number of ways
(1,3,6)	$3! = 6$
(1,4,5)	$3! = 6$
(2,2,6)	$\frac{3!}{2!} = \frac{6}{2} = 3$
(2,3,5)	$3! = 6$
(2,4,4)	$\frac{3!}{2!} = \frac{6}{2} = 3$
(3,3,4)	$\frac{3!}{2!} = \frac{6}{2} = 3$
Total	27

i.e. $n(A) = 27$

$$P(A) = \frac{n(A)}{n(S)} = \frac{27}{216} = \frac{1}{8}$$

4. Let x be one of the first hundred natural numbers chosen at random. Find the probability that $x + \frac{100}{x}$ is greater than 50

Solution:

$S = \text{first hundred natural numbers}$

$$S = \{1, 2, 3, 4, \dots, 100\}$$

$$n(S) = 100$$

$$\text{Let } A = \left\{x \mid x + \frac{100}{x} > 50\right\}$$

We have,

$$x + \frac{100}{x} > 50$$

$$x^2 + 100 > 50x$$

$$x^2 - 50x + 100 > 0$$

$$(x - 47.92)(x - 2.08) > 0 \text{ or } (47.92 - x)(2.08 - x) > 0$$

$$(x - 47.92) > 0, (x - 2.08) > 0, (47.92 - x) > 0, (2.08 - x) > 0$$

$$x > 47.92, x > 2.08, x < 47.92, x < 2.08$$

$$x = 48, 49, 50, 51, \dots, 100, x = 2, 1$$

Now,

$$A = \{1, 2, 48, 49, 50, 51, \dots, 100\}$$

$$n(A) = 55$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{55}{100} = 0.55$$



5. Each coefficient of the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary die. Find the probability that the equation will have real roots

Solution:

We have, $ax^2 + bx + c = 0$

Its formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

We know that, for the roots to be real, $b^2 - 4ac$ must be positive i.e.

$$b^2 - 4ac \geq 0$$

$$\text{i.e., } b^2 \geq 4ac$$

ac	a	c	$4ac$	b	b^2	No. of possibilities of b
1	1	1	4	2,3,4,5,6	4,9,16,25,36	5
2	1	2	8	3,4,5,6	9,16,25,36	8
	2	1		3,4,5,6	9,16,25,36	
3	1	3	12	4,5,6	16,25,36	6
	3	1		4,5,6	16,25,36	
4	1	4	16	4,5,6	16,25,36	9
	2	2		4,5,6	16,25,36	
	4	1		4,5,6	16,25,36	
5	1	5	20	5,6	25,36	4
	5	1		5,6	25,36	
6	1	6	24	5,6	25,36	8
	2	3		5,6	25,36	
	3	2		5,6	25,36	
	6	1		5,6	25,36	
7	-	-	-	-	-	-
8	2	4	32	6	36	2
	4	2		6	36	
9	3	3	36	6	36	1
Total						43

Thus, the required probability,

$$P(A) = \frac{n(A)}{n(S)} = \frac{43}{6 \times 6 \times 6} = \frac{43}{216}$$

6. Find the probability that there will be:
- (i) 53 Sundays in a non-leap year
 - (ii) 5 Sundays in February of a leap year and a non-leap year
 - (iii) 5 Sundays in the month of October
 - (iv) 5 Sundays in the month of November
 - (v) 53 Thursdays or 53 Fridays in a non-leap year

Solution:

(i) In a non-leap year there are 365 days i.e. 52 weeks and 1 day extra.

$$S = \{M, T, W, Th, F, Sa, Su\}$$

$$n(S) = 7$$

$$A = \{Su\}$$

$$n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{7}$$

(ii) in a February of a leap year, there are 29 days i.e 4 weeks and 1 day extra

$$S = \{M, T, W, Th, F, Sa, Su\}$$

$$n(S) = 7$$

$$A = \{Su\}$$

$$n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{7}$$

In a February of a non-leap year, there are 28 days i.e. 4 weeks and 0 day extra.

$$P(A) = 0$$

(iii) In the month of October, there are 31 days i.e. 4 weeks and 3 days extra

$$S = \{MTW, TWTh, WThF, ThFSa, FSaSu, SaSuM, SuMT\}$$

$$n(S) = 7$$

$$A = \{FSaSu, SaSuM, SuMT\}$$

$$n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{7}$$

(iv) In the month of November, there are 30 days i.e. 4 weeks and 2 days extra

$$S = \{MT, TW, WTh, ThF, FSa, SaSu, SuM\}$$

$$n(S) = 7$$

$$A = \{SaSu, SuM\}$$

$$n(A) = 2$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{7}$$

(v) in a non-leap year there are 365 days i.e. 52 weeks and 1 day extra

$$S = \{M, T, W, Th, F, Sa, Su\}$$



$$n(S) = 7$$

$$A = 1 \text{ Thu or } 1 \text{ Fri}$$

$$A = \{Th, F\}$$

$$n(A) = 2$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{7}$$

7. A ball is drawn at random from a box containing 12 red, 18 white, 10 blue and 15 orange balls. Find the probability that

- (i) It is red or blue
- (ii) White, blue or orange
- (iii) Neither white nor orange

Solution:

Total balls = 12 red + 18 white + 10 blue + 15 orange = 55

$S = 1 \text{ ball is drawn out of } 55$

$$n(S) = 55C1 = 55$$

nCr i.e. total n items and r is to be drawn items

- (i) let $A = a \text{ red ball or a blue ball}$

$$n(A) = 12C1 + 10C1$$

$$n(A) = 12 + 10 = 22$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{22}{55} = \frac{2}{5}$$

- (ii) let $B = a \text{ white or a blue or an orange}$

$$n(B) = 18C1 + 10C1 + 15C1$$

$$n(B) = 18 + 10 + 15 = 43$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{43}{55}$$

- (iii) let $C = \text{neither white nor orange}$

$C = \text{either blue or either red}$

$$n(C) = 10C1 + 12C1$$

$$n(C) = 10 + 12 = 22$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{22}{55} = \frac{2}{5}$$

8. There are two bags. The first contains 5 red and 7 white balls and the second contains 3 red and 12 white balls. One ball is taken out at random from the first bag and is put in the second bag. Now a ball is drawn from the second bag. What is the probability that this last ball is red?

Solution:

Bag 1 = 5 red + 7 white = 12

Bag 2 = 3 red + 12 white = 15

Let

$A = \text{red ball is transferred from bag 1 to bag 2 and a red ball is drawn out of bag 2}$



$$P(A) = \frac{5}{12} \times \frac{4}{16} = \frac{20}{192}$$

Let

B = white ball is transferred from bag 1 to bag 2 and a red ball is drawn out of bag 2

$$P(B) = \frac{7}{12} \times \frac{3}{16} = \frac{21}{192}$$

$$\text{Required probability} = P(A) + P(B) = \frac{20}{192} + \frac{21}{192} = \frac{41}{192}$$

9. The authorities of a university wish to form a committee of 4 engineers for inspecting the functioning of engineering colleges. 3 Electronics, 2 Civil, 4 Mechanical and 1 Computer Engineers showed interest to work as committee members. Find the probability that the committee consists of (i) one engineer of each branch (ii) at least one Electronics Engineer.

Solution:

$$\text{Total engineers} = 3 \text{ etrx} + 2 \text{ civil} + 4 \text{ mech} + 1 \text{ comp} = 10$$

$$S = 4 \text{ selected out of } 10$$

$$n(S) = {}^{10}C_4 = 210$$

(i) let $A = 1 E \text{ and } 1 C \text{ and } 1 M \text{ and } 1 CO$

$$n(A) = {}^3C_1 \times {}^2C_1 \times {}^4C_1 \times 1$$

$$n(A) = 3 \times 2 \times 4 \times 1 = 24$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{24}{210} = \frac{4}{35}$$

(ii) let $B = 1E \& 3 \text{others or } 2E \& 2 \text{others or } 3E \& 1 \text{other}$

$$n(B) = {}^3C_1 \times {}^7C_3 + {}^3C_2 \times {}^7C_2 + {}^3C_3 \times {}^7C_1$$

$$n(B) = 3 \times 35 + 3 \times 21 + 1 \times 7$$

$$n(B) = 175$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{175}{210} = \frac{5}{6}$$

10. A, B and C are bidding for a contract. A has exactly half the chance that B has, B in turn is $\frac{4}{5}$ as likely as C to win the contract. What is the probability for each to win the contract if the contract is to be given to one of them?

Solution:

Let the probability of C winning the contract be x

$$P(C) = x$$

$$P(B) = \frac{4}{5}x$$

$$P(A) = \frac{1}{2} \times \frac{4}{5}x = \frac{2}{5}x$$

$$\sum P(X) = 1$$

$$P(A) + P(B) + P(C) = 1$$



$$\frac{2}{5}x + \frac{4}{5}x + x = 1$$

$$\frac{11}{5}x = 1$$

$$x = \frac{5}{11}$$

$$P(A) = \frac{2}{5} \left(\frac{5}{11} \right) = \frac{2}{11}$$

$$P(B) = \frac{4}{5} \left(\frac{5}{11} \right) = \frac{4}{11}$$

$$P(C) = \frac{5}{11}$$

11. There are 11 tickets in a box bearing numbers 1 to 11. Three tickets are drawn one after the other without replacement. Find the probability that they are drawn in the order bearing (i) even, odd, even number (ii) odd, odd, even number

Solution:

$$\text{Total} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\} = 11$$

$S = 3$ tickets drawn one by one without replacement

$$n(S) = 11 \times 10 \times 9 = 990$$

(i) let $A = \text{even, odd, even}$

$$A = 5 \times 6 \times 4 = 120$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{120}{990} = \frac{4}{33}$$

(ii) let $B = \text{odd, odd, even}$

$$B = 6 \times 5 \times 5 = 150$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{150}{990} = \frac{5}{33}$$

$S = 3$ tickets drawn one by one with replacement

$$n(S) = 11 \times 11 \times 11 = 1331$$

(i) let $A = \text{even, odd, even}$

$$A = 5 \times 6 \times 5 = 150$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{150}{1331}$$

(ii) let $B = \text{odd, odd, even}$

$$B = 6 \times 6 \times 5 = 180$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{180}{1331}$$

$S = 3$ tickets drawn at random out of 11

$$n(S) = {}^{11}C_3 = 165$$

(i) let $A = \text{even, odd, even}$

$$A = 2 \text{ evens} \& 1 \text{ odd} = {}^5C_2 \times {}^6C_1 = 10 \times 6 = 60$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{60}{165} = \frac{4}{11}$$

(ii) let $B = \text{odd, odd, even}$

$$B = 2 \text{ odds} \& 1 \text{ even} = {}^6C_2 \times {}^5C_1 = 15 \times 5 = 75$$



$$P(B) = \frac{n(B)}{n(S)} = \frac{75}{165} = \frac{5}{11}$$

12. An urn A contains 8 cards bearing numbers 1 to 8. Another urn B contains 9 cards bearing numbers 1 to 9. An urn is selected at random and a card is drawn. If the card bears an odd number one more card is drawn from the same urn. If the card bears an even number the second card is drawn from the other urn. Find the probability that both cards show (i) even numbers (ii) odd numbers

Solution:

$$A = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$P(\text{even}) = \frac{4}{8} = \frac{1}{2}$$

$$P(\text{even}) = \frac{4}{9}$$

$$P(\text{odd}) = \frac{4}{8} = \frac{1}{2}$$

$$P(\text{odd}) = \frac{5}{9}$$

(i) Both cards are even

Let C = urn A is selected

$$P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{4}{9} = \frac{4}{36} = \frac{1}{9}$$

Let D = urn B is selected

$$P(D) = \frac{1}{2} \times \frac{4}{9} \times \frac{1}{2} = \frac{1}{9}$$

$$\text{Required probability} = P(C) + P(D) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

(ii) Both cards are odd

Let C = urn A is selected

$$P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{3}{7} = \frac{3}{28}$$

Let D = urn B is selected

$$P(D) = \frac{1}{2} \times \frac{5}{9} \times \frac{4}{8} = \frac{20}{144} = \frac{5}{36}$$

$$\text{Required probability} = P(C) + P(D) = \frac{3}{28} + \frac{5}{36} = \frac{31}{126}$$

(iii) one even one odd

Let C = urn A is selected and first card is even

$$P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{5}{9} = \frac{5}{36}$$

Let D = urn A is selected and first card is odd

$$P(D) = \frac{1}{2} \times \frac{1}{2} \times \frac{4}{7} = \frac{1}{7}$$

Let E = urn B is selected and first card is even

$$P(E) = \frac{1}{2} \times \frac{4}{9} \times \frac{1}{2} = \frac{1}{9}$$

Let F = urn B is selected and first card is odd

$$P(F) = \frac{1}{2} \times \frac{5}{9} \times \frac{4}{8} = \frac{5}{36}$$

$$\text{Required Probability} = P(C) + P(D) + P(E) + P(F)$$

$$\text{Required Probability} = \frac{5}{36} + \frac{1}{7} + \frac{1}{9} + \frac{5}{36} = \frac{67}{126}$$



13. The odds that a project work will be reviewed favorably by three examiners are 5 to 2, 4 to 3 and 3 to 4. What is the probability that out of three examiners the project is reviewed favorably by majority?

Solution:

$$\text{odd}(A) = 5:2$$

$$P(A) = \frac{5}{5+2} = \frac{5}{7}, P(\bar{A}) = 1 - P(A) = 1 - \frac{5}{7} = \frac{2}{7}$$

$$\text{odd}(B) = 4:3$$

$$P(B) = \frac{4}{4+3} = \frac{4}{7}, P(\bar{B}) = \frac{3}{7}$$

$$\text{odd}(C) = 3:4$$

$$P(C) = \frac{3}{3+4} = \frac{3}{7}, P(\bar{C}) = \frac{4}{7}$$

$$P(\text{all three approves}) = P(A)P(B)P(C) = \frac{5}{7} \cdot \frac{4}{7} \cdot \frac{3}{7} = \frac{60}{343}$$

$$P(AB \text{ approves}) = P(A)P(B)P(\bar{C}) = \frac{5}{7} \cdot \frac{4}{7} \cdot \frac{4}{7} = \frac{80}{343}$$

$$P(BC \text{ approves}) = P(\bar{A})P(B)P(C) = \frac{2}{7} \cdot \frac{4}{7} \cdot \frac{3}{7} = \frac{24}{343}$$

$$P(AC \text{ approves}) = P(A)P(\bar{B})P(C) = \frac{5}{7} \cdot \frac{3}{7} \cdot \frac{3}{7} = \frac{45}{343}$$

$$P(\text{majority}) = \frac{60+80+24+45}{343} = \frac{209}{343}$$

14. In a shooting competition the probability of A hitting the target is $\frac{1}{2}$, of B is $\frac{2}{3}$ and of C is $\frac{3}{4}$. If all of them fire at the target, find the probability that (i) none of them hits the target (ii) at least one of them hit the target

Solution:

$$P(A) = \frac{1}{2}, P(\bar{A}) = 1 - P(A) = \frac{1}{2}$$

$$P(B) = \frac{2}{3}, P(\bar{B}) = 1 - P(B) = \frac{1}{3}$$

$$P(C) = \frac{3}{4}, P(\bar{C}) = 1 - P(C) = \frac{1}{4}$$

$$(i) P(\text{none}) = P(\bar{A})P(\bar{B})P(\bar{C}) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24}$$

$$(ii) P(\text{atleast 1}) = P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C) + P(A)P(B)P(\bar{C}) + P(A)P(\bar{B})P(C) + P(\bar{A})P(B)P(C) + P(A)P(B)P(C)$$

$$P(\text{atleast 1}) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4}$$

$$P(\text{atleast 1}) = \frac{23}{24}$$

OR

$$P(\text{atleast 1}) = 1 - P(\text{none})$$

$$P(\text{atleast 1}) = 1 - \frac{1}{24} = \frac{23}{24}$$



15. In a certain college 4% of the boys and 1% of the girls are taller than 1.8 mts. Furthermore 60% of the students are girls. If a student selected at random is taller than 1.8 mts, what is the probability that the student was a boy? Justify your answer.

Solution:

Let the total number of students be 1000

	Boys (A)	Girls (B)	Total
Taller than 1.8 mts (C)	16	6	22
Shorter than 1.8 mts (D)	384	594	978
Total	400	600	1000

$P(\text{student was a boy given that he is taller than 1.8 mts})$

$$P\left(\frac{A}{C}\right) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{n(A \cap C)}{n(S)}}{\frac{n(C)}{n(S)}} = \frac{n(A \cap C)}{n(C)} = \frac{16}{22} = \frac{8}{11}$$

16. There are 100 students in a class of which 36 are boys studying Statistics and 13 girls are not studying Statistics. If there are 55 girls in all, find the probability that a student selected at random is not studying Statistics given that the student is a boy.

Solution:

	Boys (A)	Girls (B)	Total
Studying stats (C)	36	42	78
Not studying stats (D)	9	13	22
Total	45	55	100

$P(\text{not studying stats given that he is a boy})$

$$P\left(\frac{D}{A}\right) = \frac{P(D \cap A)}{P(A)} = \frac{n(D \cap A)}{n(A)} = \frac{9}{45} = \frac{1}{5}$$

17. A and B toss a fair coin simultaneously. A starts. Find their probabilities of winning the toss.

Solution:

$$P(\text{head}) = \frac{1}{2}, P(\text{tail}) = \frac{1}{2}$$

$$\text{Probability of A winning} = A + \bar{A}\bar{B}A + \bar{A}\bar{B}\bar{A}BA + \dots$$

$$= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \dots$$

$$= \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \dots$$

$$= \frac{1}{2} \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \right]$$

$$= \frac{1}{2} \left[1 + \left(\frac{1}{4}\right) + \left(\frac{1}{4}\right)^2 + \dots \right]$$

$$(1 + x + x^2 + x^3 + \dots) = [1 - x]^{-1}$$



$$\begin{aligned}
 &= \frac{1}{2} \left[1 - \frac{1}{4} \right]^{-1} \\
 &= \frac{1}{2} \left[\frac{3}{4} \right]^{-1} \\
 &= \frac{1}{2} \left[\frac{4}{3} \right] \\
 &= \frac{2}{3}
 \end{aligned}$$

Thus,

$$\text{Probability of B winning} = 1 - \frac{2}{3} = \frac{1}{3}$$

18. A and B alternately throw a pair of dice. One who throws 9 first wins. If A starts, show that their chances of winning are 9 : 8.

Solution:

S = two pair of dice are rolled

$$n(S) = 36$$

W = sum is 9

$$W = \{(3,6), (4,5), (5,4), (6,3)\}$$

$$n(W) = 4$$

$$P(\text{winning}) = P(W) = \frac{n(W)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

$$P(\text{losing}) = 1 - \frac{1}{9} = \frac{8}{9}$$

$$\text{Probability of A winning} = A + \bar{A} \bar{B} A + \bar{A} \bar{B} A \bar{B} A + \dots \dots \dots$$

$$= \frac{1}{9} + \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{1}{9} + \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{8}{9} \cdot \frac{1}{9} + \dots \dots \dots$$

$$= \frac{1}{9} \left[1 + \frac{64}{81} + \left(\frac{64}{81} \right)^2 + \dots \dots \dots \right]$$

$$= \frac{1}{9} \left[1 - \frac{64}{81} \right]^{-1}$$

$$= \frac{1}{9} \left[\frac{17}{81} \right]^{-1}$$

$$= \frac{1}{9} \left[\frac{81}{17} \right]$$

$$= \frac{9}{17}$$

Thus,

$$\text{Probability of B winning} = 1 - \frac{9}{17} = \frac{8}{17}$$

$$\text{Ratio of their probabilities} = \frac{\frac{9}{17}}{\frac{8}{17}} = \frac{9}{8} = 9 : 8$$

19. A and B alternately throw a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins find their chances of winning the game

Solution:



S = pair of dice is thrown

$$n(S) = 36$$

A = sum is 6

$$A = \{(1,5), (2,4), (3,3), (4,2), (5,1)\}$$

$$n(A) = 5$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36}, P(\bar{A}) = 1 - \frac{5}{36} = \frac{31}{36}$$

B = sum is 7

$$B = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$n(B) = 6$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{6}{36}, P(\bar{B}) = 1 - \frac{6}{36} = \frac{30}{36}$$

$$\begin{aligned} \text{Probability of A winning} &= A + \bar{A} \bar{B} A + \bar{A} \bar{B} \bar{A} \bar{B} A + \dots \dots \dots \\ &= \frac{5}{36} + \frac{31}{36} \cdot \frac{30}{36} \cdot \frac{5}{36} + \frac{31}{36} \cdot \frac{30}{36} \cdot \frac{31}{36} \cdot \frac{30}{36} \cdot \frac{5}{36} + \dots \dots \dots \\ &= \frac{5}{36} \left[1 + \frac{155}{216} + \left(\frac{155}{216} \right)^2 + \dots \dots \dots \right] \\ &= \frac{5}{36} \left[1 - \frac{155}{216} \right]^{-1} \\ &= \frac{5}{36} \left[\frac{61}{216} \right]^{-1} \\ &= \frac{5}{36} \left[\frac{216}{61} \right] \\ &= \frac{30}{61} \end{aligned}$$

Thus,

$$\text{Probability of B winning} = 1 - \frac{30}{61} = \frac{31}{61}$$

20. A, B, C throw a fire die in that order. A starts. One who throws an ace first wins. Find the probabilities of their winning

Solution:

S = a die is thrown

$$n(S) = 6$$

W = an ace

$$n(W) = 1$$

$$P(W) = \frac{n(W)}{n(S)} = \frac{1}{6}$$

$$P(L) = 1 - \frac{1}{6} = \frac{5}{6}$$

$$\begin{aligned} \text{Probability of A winning} &= A + \bar{A} \bar{B} \bar{C} A + \bar{A} \bar{B} \bar{C} \bar{A} \bar{B} \bar{C} A + \dots \dots \dots \\ &= \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \dots \dots \dots \\ &= \frac{1}{6} \left[1 + \frac{125}{216} + \left(\frac{125}{216} \right)^2 + \dots \dots \dots \right] \\ &= \frac{1}{6} \left[1 - \frac{125}{216} \right]^{-1} \end{aligned}$$



$$= \frac{1}{6} \left[\frac{91}{216} \right]^{-1}$$

$$= \frac{1}{6} \left[\frac{216}{91} \right]$$

$$= \frac{36}{91}$$

$$\text{Probability of B winning} = \overline{A}B + \overline{A}\overline{B}\overline{C}\overline{A}B + \overline{A}\overline{B}\overline{C}\overline{A}\overline{B}\overline{C}\overline{A}B + \dots$$

$$= \frac{5}{6} \cdot \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \dots$$

$$= \frac{5}{6} \cdot \frac{1}{6} \left[1 + \frac{125}{216} + \left(\frac{125}{216} \right)^2 + \dots \right]$$

$$= \frac{5}{36} \left[1 - \frac{125}{216} \right]^{-1}$$

$$= \frac{5}{36} \left[\frac{91}{216} \right]^{-1}$$

$$= \frac{5}{36} \left[\frac{216}{91} \right]$$

$$= \frac{30}{91}$$

$$\text{Probability of C winning} = 1 - \left(\frac{36}{91} + \frac{30}{91} \right) = \frac{25}{91}$$

21. If three dice are thrown, find the probability that the total score is 12

$$\text{Ans. } \frac{25}{216}$$

22. What is the chance that a leap year selected at random will contain 53 Sundays?

$$\text{Ans. } \frac{2}{7}$$

23. A can hit a target 2 times in 5 shots, B 3 times in 4 shots, C 2 times in 3 shots. They fire a volley. What is the probability that at least 2 shots hit the target?

$$\text{Ans. } \frac{2}{3}$$

24. The odds against A speaking the truth are 4:6, while the odds in favor of B speaking the truth are 7:3. What is the probability that A and B contradict each other in stating the same fact?

$$\text{Ans. } \frac{23}{50}$$

25. A card is drawn from a pack of cards. Find the probability that it will be (i) a picture card given that it is a spade (ii) a spade given that it is a picture card

$$\text{Ans. } \frac{3}{13}, \frac{1}{4}$$

26. Out of one lakh people 51500 are female and 48500 are male. Among the females 9000 are singers, among the males 30200 are singers. A person chosen randomly. If A, B, C are the events that a singer is chosen, a female is chosen and a male is chosen respectively then find

(i) $P(A/B)$ (ii) $P(A/C)$ (iii) $P(B/A)$ (iv) $P(C/A)$

$$\text{Ans. } 0.1748, 0.6227, 0.2296, 0.7704$$

27. Given that a student had prepared, the probability of passing a certain entrance exam is 0.99. Given that a student did not prepare, the



probability of passing the entrance exam is 0.05. Assume that the probability of preparing is 0.7. The student fails in the exam. What is the probability that he or she did not prepare? Ans. 0.976

28. In a factory there are 100 workers as shown below:

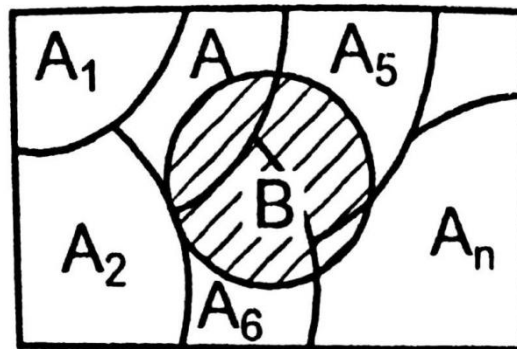
	Male	Female	Total
Skilled	35	25	60
Unskilled	30	10	40
Total	65	35	100

Find the probability that a worker selected at random will be (i) a male given that he is skilled (ii) unskilled given that she is female

Ans. $\frac{7}{12}, \frac{2}{7}$

29. A and B alternately throw a pair of dice. One who throws 10 first wins. If A starts, show that their chances of winning are 12 : 11.

Type II: Total Probability Theorem & Baye's Theorem



Total Probability Theorem

Let $A_1, A_2, A_3, \dots, A_n$ be a partition of S and B be some event defined on S. then,

$$P(B) = P(B/A_1) \times P(A_1) + P(B/A_2) \times P(A_2) + \dots$$

If we denote the probabilities $P(A_i)$ as p_i and the conditional probabilities $P(B/A_i)$ as p'_i then the above theorem can be stated as

$$P(B) = p_1 p'_1 + p_2 p'_2 + p_3 p'_3 + \dots + p_n p'_n$$

The total probability of B

Bayes Theorem

Let the events $A_1, A_2, \dots, A_n$ represent a partition of the sample space S. Let B be any other event defined on S. if $P(A_i) \neq 0, i = 1, 2, \dots, n$ and $P(B) \neq 0$ then



$$P(A_i/B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(A_i) \times P(B/A_i)}{P(B)}$$

If we denote the probabilities $P(A_i)$ as p_i and the conditional probabilities $P(B/A_i)$ as p'_i then the above theorem can be stated as

It can also be stated as,

$$P(A_i/B) = \frac{p_i p'_i}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3 + \dots + p_n p'_n}$$

1. There are in a bag three true coins and one false coin with head on both sides. A coin is chosen at random and tossed four times. If head occurs all the four times, what is the probability that the false coin was chosen and used?

Solution:

$$\text{Bag} = \{3 \text{ true \& 1 False}\} = 4$$

$$P(\text{selecting true coin}) = p_1 = \frac{3}{4}$$

$$P(\text{selecting false coin}) = p_2 = \frac{1}{4}$$

$$p'_1 = P(\text{getting all four heads with true coin}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$$

$$p'_2 = P(\text{getting all four heads with false coin}) = 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

$$\text{Total probability} = p_1 p'_1 + p_2 p'_2 = \frac{3}{4} \cdot \frac{1}{16} + \frac{1}{4} \cdot 1 =$$

$$\text{Required Probability} = \frac{p_2 p'_2}{p_1 p'_1 + p_2 p'_2} = \frac{\frac{1}{4} \cdot 1}{\left(\frac{3}{4}\right)\left(\frac{1}{16}\right) + \left(\frac{1}{4}\right)(1)} = \frac{16}{19}$$

2. A coin is tossed. If it turns up heads two balls are drawn from urn A otherwise two balls are drawn from urn B. Urn A contains 3 black and 5 white balls. Urn B contains 7 black and one white ball. What is the probability that urn A was used, given that both balls drawn are black?

Solution:

$$p_1 = P(H) = \frac{1}{2}$$

$$p_2 = P(T) = \frac{1}{2}$$

$$p'_1 = P(\text{two black balls from A}) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$$

$$p'_2 = P(\text{two black balls from B}) = \frac{{}^7C_2}{{}^8C_2} = \frac{3}{4}$$

$$\text{Required Probability} = \frac{p_1 p'_1}{p_1 p'_1 + p_2 p'_2} = \frac{\left(\frac{1}{2}\right)\left(\frac{3}{28}\right)}{\left(\frac{1}{2}\right)\left(\frac{3}{28}\right) + \left(\frac{1}{2}\right)\left(\frac{3}{4}\right)} = \frac{1}{8}$$



3. A bag contains 7 red and 3 black balls and another bag contains 4 red and 5 black balls. One ball is transferred from the first bag to the second bag and then a ball is drawn from the second bag. If this ball happens to be red, find the probability that a black ball was transferred.

Solution:

$$p_1 = \text{Probability of transferring black ball} = \frac{3}{10}$$

$$p_2 = \text{Probability of transferring red ball} = \frac{7}{10}$$

$$p'_1 = \text{Probability of drawing a red ball if black was transferred} = \frac{4}{10}$$

$$p'_2 = \text{Probability of drawing a red ball if red was transferred} = \frac{5}{10}$$

$$\text{Required Probability} = \frac{p_1 p'_1}{p_1 p'_1 + p_2 p'_2} = \frac{\left(\frac{3}{10}\right)\left(\frac{4}{10}\right)}{\left(\frac{3}{10}\right)\left(\frac{4}{10}\right) + \left(\frac{7}{10}\right)\left(\frac{5}{10}\right)} = \frac{12}{47}$$

4. In a certain test there are multiple choice questions. There are four possible answers to each question and one of them is correct. An intelligent student can solve 90% questions correctly by reasoning and for the remaining 10% questions he gives answers by guessing. A weak student can solve 20% questions correctly by reasoning and for the remaining 80% questions he gives answers by guessing. An intelligent students gets the correct answer, what is the probability that he was guessing?

[M19/IT/8M]

Solution:

Consider the intelligent student,

$$p_1 = \text{answer by reasoning} = \frac{90}{100} = \frac{9}{10}$$

$$p_2 = \text{answer by guessing} = \frac{10}{100} = \frac{1}{10}$$

$$p'_1 = \text{answer is correct by reasoning} = 1$$

$$p'_2 = \text{answer is correct by guessing} = \frac{1}{4}$$

$$\text{Required Probability} = \frac{p_2 p'_2}{p_1 p'_1 + p_2 p'_2} = \frac{\left(\frac{1}{10}\right)\left(\frac{1}{4}\right)}{\left(\frac{9}{10}\right)(1) + \left(\frac{1}{10}\right)\left(\frac{1}{4}\right)} = \frac{1}{37}$$

5. A man speaks truth 3 times out of 5. When a die is thrown, he states that it gave an ace. What is the probability that this event has actually happened?

[N19/IT/8M]

Solution:

$$p_1 = \text{prob of truth} = \frac{3}{5}, p_2 = \text{prob of lie} = \frac{2}{5}$$



$$p'_1 = \text{prob of ace} = \frac{1}{6}, p'_2 = \text{prob of not ace} = \frac{5}{6}$$

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2} = \frac{\left(\frac{3}{5}\right)\left(\frac{1}{6}\right)}{\left(\frac{3}{5}\right)\left(\frac{1}{6}\right) + \left(\frac{2}{5}\right)\left(\frac{5}{6}\right)} = \frac{3}{13}$$

6. A lot of IC chips is known to contain 3% of defective chips. Each chip is tested before delivery but the tester is not completely reliable. It is known that:

$P(\text{Tester says the chip is good}/\text{The chip is actually good})=0.95$

$P(\text{Tester says the chip is defective}/\text{The chip is actually defective})=0.96$

If a tested chip is declared defective by the tester. What is the probability that it is actually defective?

Solution:

$p_1 = \text{chip is defective} = 0.03$

$p'_1 = \text{tester says defective and it is defective} = 0.96$

$p_2 = \text{chip is not defective} = 0.97$

$p'_2 = \text{tester says defective and it is not defective} = 0.05$

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2} = \frac{(0.03)(0.96)}{(0.03)(0.96) + (0.97)(0.05)} = 0.37$$

7. A certain test for particular cancer is known to be 95% accurate. A person submits to the test and result is positive. Suppose that a person comes from a population of 1,00,000 where 2000 people suffer from disease. What can we conclude about the probability that person under test has particular cancer?

[N17/IT/6M]

Solution:

$p_1 = \text{prob a person has cancer} = \frac{2000}{100000} = \frac{2}{100} = 0.02$

$p_2 = \text{prob a person does not has cancer} = 1 - 0.02 = 0.98$

$p'_1 = \text{test is positive when a person has cancer} = \frac{95}{100} = 0.95$

$p'_2 = \text{test is positive when a person does not has cancer} = 0.05$

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2} = \frac{(0.02)(0.95)}{(0.02)(0.95) + (0.98)(0.05)} = \frac{19}{68}$$

8. In a bolt factory, machines A,B, C produce respectively 25%, 35% and 40%. Of their output 5%, 4% and 2% are defective. A bolt is drawn at random from a days production and is found defective. What is the probability that it was produced by machines A, B, C?

[M18/IT/8M][N18/IT/8M]

Solution:

$p_1 = \text{produced by A} = 0.25$



$$p_2 = \text{produced by B} = 0.35$$

$$p_3 = \text{produced by C} = 0.40$$

$$p'_1 = \text{defective by A} = 0.05$$

$$p'_2 = \text{defective by B} = 0.04$$

$$p'_3 = \text{defective by C} = 0.02$$

Defective obtained was produced by machine A,

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{25}{69}$$

Defective obtained was produced by machine B,

$$\text{Required Probability} = \frac{p_2 \cdot p'_2}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{28}{69}$$

Defective obtained was produced by machine C,

$$\text{Required Probability} = \frac{p_3 \cdot p'_3}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{16}{69}$$

9. Three factories A,B, C produces 30%, 50% and 20% of the total production of an item. Out of their production 8%, 5% and 1% are defective. Find the probability that defective item is produced by factory A?

[N18/Elect/6M]

Solution:

$$p_1 = \text{produced by A} = 0.30$$

$$p_2 = \text{produced by B} = 0.50$$

$$p_3 = \text{produced by C} = 0.20$$

$$p'_1 = \text{defective by A} = 0.08$$

$$p'_2 = \text{defective by B} = 0.05$$

$$p'_3 = \text{defective by C} = 0.01$$

Defective obtained was produced by factory A,

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{8}{17}$$

10. In a factory machine A,B, C produces 30%, 50% and 20% of the total production of an item. Out of their production 80%, 50% and 10% are defective. Find the probability that defective item is produced by machine A?

[M19/Extc/6M]

Solution:

$$p_1 = \text{produced by A} = 0.30$$

$$p_2 = \text{produced by B} = 0.50$$

$$p_3 = \text{produced by C} = 0.20$$

$$p'_1 = \text{defective by A} = 0.8$$

$$p'_2 = \text{defective by B} = 0.5$$

$$p'_3 = \text{defective by C} = 0.1$$



Defective obtained was produced by factory A,

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{8}{17}$$

11. A box contains three biased coins A, B and C. the probability that a head will result when A is tossed is $\frac{1}{3}$, when B is tossed is $\frac{2}{3}$ and when C is tossed it is $\frac{3}{4}$. If one of the coins is chosen and is tossed 3 times, head resulted twice and tail once. What is the probability that the coin chosen was A?

Solution:

$$p_1 = P(\text{choosing A}) = \frac{1}{3}$$

$$p_2 = P(\text{choosing B}) = \frac{1}{3}$$

$$p_3 = P(\text{choosing C}) = \frac{1}{3}$$

$$p'_1 = P(\text{getting 2 heads in 3 tosses with A}) = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = \frac{2}{9}$$

$$p'_2 = P(\text{getting 2 heads in 3 tosses with B}) = {}^3C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right) = \frac{4}{9}$$

$$p'_3 = P(\text{getting 2 heads in 3 tosses with C}) = {}^3C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right) = \frac{27}{64}$$

$$\text{Required Probability} = \frac{p_1 \cdot p'_1}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3} = \frac{128}{627}$$

12. A bag contains five balls, the colors of which are not known. Two balls were drawn from the bag and they were found to be white. What is the probability that all balls are white?

Solution:

Since two balls drawn are white, the bag may contain 2 white or 3 white or 4 white or 5 white balls.

$$p_1 = P(\text{bag contains 2 whites}) = \frac{1}{4}$$

$$p_2 = P(\text{bag contains 3 whites}) = \frac{1}{4}$$

$$p_3 = P(\text{bag contains 4 whites}) = \frac{1}{4}$$

$$p_4 = P(\text{bag contains 5 whites}) = \frac{1}{4}$$

$$p'_1 = P(\text{drawing 2 balls white when 2 are white}) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}$$

$$p'_2 = P(\text{drawing 2 balls white when 3 are white}) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10}$$

$$p'_3 = P(\text{drawing 2 balls white when 4 are white}) = \frac{{}^4C_2}{{}^5C_2} = \frac{3}{5}$$

$$p'_4 = P(\text{drawing 2 balls white when 5 are white}) = \frac{{}^5C_2}{{}^5C_2} = 1$$

$$\text{Required Probability} = \frac{p_4 p'_4}{p_1 p'_1 + p_2 p'_2 + p_3 p'_3 + p_4 p'_4} = \frac{1}{2}$$

